

[6004]-616

B.E. (Mechanical Engineering)
Computer Integrated Manufacturing
(2019 Pattern) (Semester - VIII) (402048)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q. 7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Use of electronic pocket calculator is allowed.
- 5) Assume suitable data, if necessary.

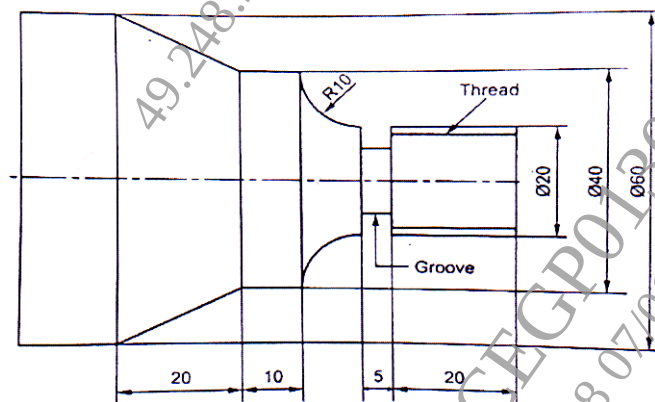
Q1) a) Define Computer aided Manufacturing (CAM). Explain CAM with its objectives & benefits to industry. [6]

b) Write a complete part program using G code & M code for given job Figure 1b as below. Assume suitable data and feed for machining. [12]

Billet size – Dia. 60mm & length – 90mm

Thread – Do = 20mm, Dc = 17mm & pitch = 2.5mm.

Groove – width – 5mm & depth – 5mm.

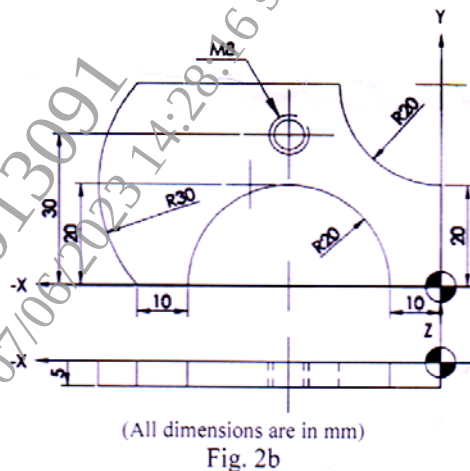


(All dimensions are in mm)

Fig. 1b

OR

- Q2) a)** Differentiate between Numerical control Machine (NC) & Computerized Numerical control Machine (CNC). [6]
- b)** Write a CNC program for the part shown in Figure 2b. Assume suitable data. [12]



- Q3) a)** Explain Computer Aided Process Planning (CAPP) and its Benefits. [8]
- b)** Explain the importance of control system in automated production system. Write the concept in short Inventory & Shop floor control. [9]

OR

- Q4) a)** Write a note on Material Requirement Planning with input, working, outputs and benefits. [8]
- b)** Explain Computer aided inspection & quality control. Explain any two methods of computer aided inspection technique. [9]

- Q5) a)** Explain with neat sketch concept of Flexible Manufacturing System (FMS). Write Objectives & area of application. [6]
- b)** Consider a condition of 5 machines and 10 parts. Create group of machines by using Rank order clustering method. [12]

Machines	1	2	3	4	5	6	7	8	9	10
M1	1	1	1	1	1		1	1	1	1
M2		1	1	1					1	1
M3	1				1	1	1			
M4		1	1	1				1	1	1
M5	1	1	1	1		1	1	1		

OR

- Q6)** a) What is Group Technology? Explain the part classification and Opitz Coding system. [6]
- b) Five machines will constitute a GT cell. The from to data for the machines are shown in the table below. Determine the most logical sequence of machines for this data using hollier method. [12]

From :	1	2	3	4	5
1	0	10	80	0	0
2	0	0	0	85	0
3	0	0	0	0	0
4	70	0	20	0	0
5	0	75	0	20	0

- Q7)** a) What is Internet of Things (IoT)? Explain & illustrate the components of Internet of Things (IoT). [8]
- b) What are the aspects of Digital manufacturing? Explain features and any five benefits of Digital Manufacturing. [9]
- Q8)** a) What is Industry 4.0? Explain the functions of components of Industry 4.0. [8]
- b) Explain and illustrate Cyber-Physical Manufacturing Systems with Features of Cyber physical system (CPS). [9]
